

# Supplementary Information: PRISMA 2020 Checklist

*Financial Time-Series Generation: A Systematic Survey of Taxonomy, Evaluation, Datasets, and Open Challenges*

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This document provides the completed **PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) Checklist** demonstrating methodological compliance of our systematic survey with established reporting standards.

| Section/Topic                 | Item # | Checklist Item                                                                                                                                                                    | Location in Manuscript                                      |
|-------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>TITLE</b>                  | 1      | Identify the report as a systematic review.                                                                                                                                       | Title page (Title contains “A Systematic Survey”)           |
| <b>ABSTRACT</b>               | 2      | See the PRISMA 2020 for Abstracts checklist. Provide structured/unstructured summary of background, objectives, eligibility criteria, synthesis methods, results, and discussion. | Abstract (single paragraph, 181 words, Page 1)              |
| <b>INTRODUCTION</b>           |        |                                                                                                                                                                                   |                                                             |
| Rationale                     | 3      | Describe the rationale for the review in the context of what is already known.                                                                                                    | Introduction (Paragraphs 1–3)                               |
| Objectives                    | 4      | Provide an explicit statement of the questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design.                            | Introduction (Paragraphs 4–7, Table 1)                      |
| <b>METHODS</b>                |        |                                                                                                                                                                                   |                                                             |
| Eligibility criteria          | 5      | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.                                                                       | Methods (Inclusion and Exclusion Criteria)                  |
| Information sources           | 6      | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted.                                                               | Methods (Literature Retrieval Strategy)                     |
| Search strategy               | 7      | Present the full search strategies for all databases, registers and websites, including any filters and limits used.                                                              | Methods (Literature Retrieval Strategy)                     |
| Selection process             | 8      | State the processes for deciding which studies were eligible, including how many reviewers screened each record and whether they worked independently.                            | Methods (Inclusion and Exclusion Criteria, Fig. 6)          |
| Data collection process       | 9      | State the methods of extracting data from reports and how reviewers confirmed data.                                                                                               | Methods (Data Extraction and Coding Protocol)               |
| Data items                    | 10a    | List and define all outcomes for which data were sought.                                                                                                                          | Methods (Data Extraction and Coding Protocol)               |
|                               | 10b    | List and define all other variables sought (e.g. participant/intervention characteristics, funding sources).                                                                      | Methods (Data Extraction and Coding Protocol)               |
| Study risk of bias assessment | 11     | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used.                                                                   | Methods (Inclusion and Exclusion Criteria, Data Extraction) |
| Effect measures               | 12     | Specify for each outcome the effect measure(s) used in the synthesis or presentation of results.                                                                                  | Results (Evaluation Protocols)                              |

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|-------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Synthesis methods             | 13a–13f | Describe the processes used to decide which studies were eligible for each synthesis, data preparation, tabular display methods, and synthesis of results. | Methods (Review Dimensions and Taxonomy Synthesis, Fig. 6)                         |
| Reporting bias assessment     | 14      | Describe any methods used to assess the risk of bias due to missing results in a synthesis.                                                                | Results (Data Resources and Benchmarks; dataset construction pitfalls)             |
| Certainty assessment          | 15      | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.                                                      | Results (Evaluation Protocols)                                                     |
| <b>RESULTS</b>                |         |                                                                                                                                                            |                                                                                    |
| Study selection               | 16a     | Describe the results of the search and selection process, from the number of records identified to the number of studies included.                         | Methods (Fig. 6: PRISMA screening pipeline; Table 8: retrieval stats)              |
|                               | 16b     | Cite studies that appeared to meet the inclusion criteria but were excluded, and explain why.                                                              | Methods (Table 8 footnote and exclusion list)                                      |
| Study characteristics         | 17      | Cite each included study and present its characteristics.                                                                                                  | Results (Extrapolation, Imputation, Synthesis; Tables 1–6)                         |
| Risk of bias in studies       | 18      | Present assessments of risk of bias for each included study.                                                                                               | Results (Data Resources and Benchmarks; dataset construction pitfalls)             |
| Results of individual studies | 19      | For all outcomes, present for each study summary statistics or findings.                                                                                   | Results (Taxonomy mappings and Comparative Tables)                                 |
| Results of syntheses          | 20a–20d | Present the results of the syntheses, including characteristics of included models, cross-paradigm comparisons, and visual representations.                | Results (Taxonomy Tree Fig. 2, Task Flow Fig. 3, Comparisons Tables 3–5)           |
| Reporting biases              | 21      | Present assessments of risk of bias due to missing results (survivorship bias, look-ahead bias).                                                           | Results (Data Resources and Benchmarks; dataset construction pitfalls)             |
| Certainty of evidence         | 22      | Present assessments of certainty in the body of evidence.                                                                                                  | Results (closing paragraphs of the Extrapolation, Imputation, and Synthesis parts) |
| <b>DISCUSSION</b>             |         |                                                                                                                                                            |                                                                                    |
| Discussion                    | 23a     | Provide a general interpretation of the results in the context of other evidence.                                                                          | Discussion (Paragraph 1)                                                           |
|                               | 23b     | Discuss any limitations of the evidence included in the review.                                                                                            | Discussion (limitations and practical implications)                                |
|                               | 23c     | Discuss any limitations of the review processes used.                                                                                                      | Discussion (Limitations, practical implications, and concluding outlook)           |
|                               | 23d     | Discuss implications of the results for practice, policy, and future research.                                                                             | Discussion (Causal learning, Econometric bounds, Multimodality, Online adaptation) |
| <b>OTHER INFORMATION</b>      |         |                                                                                                                                                            |                                                                                    |

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|----------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Registration and protocol  | 24a–24c       | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | Methods Retrieval Strategy & PRISMA compliance statement)         |
| Support                    | 25            | Describe sources of financial or non-financial support for the review, and the role of the funders.                                            | Declarations (Funding statement)                                  |
| Competing interests        | 26            | Declare any competing interests of review authors.                                                                                             | Declarations (Competing Interests statement)                      |
| Availability of data, code | 27            | Report which of the following are publicly available and where they can be found: data, code, extraction forms, logs.                          | Declarations (Data Availability and Code Availability statements) |